

Homework 03 - Loops, Classes And Objects

Topics Covered

- `while` , `do-while` , and `for`
- Nested loops
- `break` and `continue`
- Class and object
- Fields/instance variables
- Creating objects using `new`
- Accessing fields from another class using the dot operator

Instructions

1. Write output answers in your notebook.
2. Write each programming question in a separate Java file.
3. Use meaningful class and variable names.
4. Dry-run loop questions before running them.
5. For class/object programs, create the object inside `main` .

Part A: Short Questions

1. What is a loop?
2. What are the three main parts of a loop?
3. When should we use a `for` loop?
4. When should we use a `while` loop?
5. Which loop executes at least once?
6. What is an infinite loop?
7. What does `break` do?
8. What does `continue` do?
9. What is a nested loop?
10. What is a class?
11. What is an object?
12. What is a field or instance variable?
13. What does the `new` keyword do?
14. What is a reference variable?
15. What is the dot operator used for?

Part B: Find The Output

Question 1

```
for (int i = 1; i <= 5; i++) {  
    System.out.print(i + " ");  
}
```

Question 2

```
for (int i = 10; i >= 2; i -= 2) {  
    System.out.print(i + " ");  
}
```

Question 3

```
int i = 1;  
  
while (i <= 5) {  
    System.out.print(i * i + " ");  
    i++;  
}
```

Question 4

```
int i = 10;  
  
do {  
    System.out.println(i);  
    i++;  
} while (i < 5);
```

Question 5

```
for (int i = 1; i <= 6; i++) {  
    if (i == 4) {  
        continue;  
    }  
  
    System.out.print(i + " ");  
}
```

Question 6

```
for (int i = 1; i <= 10; i++) {  
    if (i == 5) {  
        break;  
    }  
  
    System.out.print(i + " ");  
}
```

Question 7

```
int sum = 0;

for (int i = 1; i <= 5; i++) {
    sum += i;
}

System.out.println(sum);
```

Question 8

```
for (int row = 1; row <= 3; row++) {
    for (int column = 1; column <= 2; column++) {
        System.out.print("* ");
    }

    System.out.println();
}
```

Question 9

```
class Student {
    String name;
    int age;
}

public class Main {
    public static void main(String[] args) {
        Student student = new Student();

        student.name = "Rahul";
        student.age = 20;

        System.out.println(student.name);
        System.out.println(student.age);
    }
}
```

Question 10

```
class Product {
    String name;
    double price;
}

public class Main {
    public static void main(String[] args) {
        Product first = new Product();
        Product second = new Product();

        first.name = "Laptop";
        first.price = 50000;

        second.name = "Mouse";
        second.price = 500;

        first.price = 48000;

        System.out.println(first.price);
        System.out.println(second.price);
    }
}
```

Part C: Find And Correct The Error

Question 1

```
for (int i = 1; i <= 5; i--) {
    System.out.println(i);
}
```

Question 2

```
int i = 1;

while (i <= 5) {
    System.out.println(i);
}
```

Question 3

```
for (int i = 0; i < 5; i++);
{
    System.out.println("Java");
}
```

Question 4

```

class Student {
    String name;
}

public class Main {
    public static void main(String[] args) {
        Student student;
        student.name = "Rahul";
    }
}

```

Question 5

```

class Car {
    String color;
}

public class Main {
    public static void main(String[] args) {
        Car car = new Car();
        color = "Red";

        System.out.println(car.color);
    }
}

```

Part D: Loop Programs

Program 1: Number Series

Print numbers from 1 to 20 using a `for` loop.

Program 2: Reverse Series

Print numbers from 20 to 1 using a `while` loop.

Program 3: Even And Odd Numbers

Print:

- even numbers from 1 to 50
- odd numbers from 1 to 50

Program 4: Multiplication Table

Create a variable:

```
int number = 7;
```

Print its multiplication table from 1 to 10.

Program 5: Sum Of Numbers

Find the sum of numbers from 1 to 100.

Program 6: Factorial

Find the factorial of 5.

Expected result:

```
120
```

Program 7: Divisible Numbers

Print numbers from 1 to 100 that are divisible by both 3 and 5.

Program 8: Break And Continue

Print numbers from 1 to 20:

- skip 7 using continue
- stop the loop when the number becomes 15

Program 9: Square Pattern

Print:

```
* * * *
* * * *
* * * *
* * * *
```

Program 10: Increasing Pattern

Print:

```
*
* *
* * *
* * * *
* * * * *
```

Part E: Class And Object Programs

Program 1: Student

Create a Student class containing:

```
String name;
int rollNumber;
int age;
int marks;
```

In another class containing main :

1. Create a Student object.
2. Assign values using the dot operator.
3. Print all values.

Program 2: Two Students

Create two Student objects.

Store different names, roll numbers, and marks in both objects.

Print both student records.

Program 3: Car

Create a Car class containing:

```
String brand;  
String color;  
double price;
```

Create two Car objects in another class and print their details.

Program 4: Product

Create a Product class containing:

```
String name;  
double price;  
int quantity;
```

Create an object and calculate:

```
total price = price * quantity
```

Program 5: Employee

Create an Employee class containing:

```
int employeeId;  
String employeeName;  
double monthlySalary;
```

Create an object and calculate annual salary.

Program 6: Bank Account

Create a BankAccount class containing:

```
String accountHolder;  
String accountNumber;  
double balance;
```

Create an object, assign values, and print account details.

Part F: Combined Logic Challenges

Challenge 1: Store Multiple Students

Create a Student class.

Inside `main` , create three `Student` objects and assign different values.

Print the student who has the highest marks using `if-else` .

Challenge 2: Product Stock

Create a `Product` class with:

```
String name;  
double price;  
int stock;
```

Create three `Product` objects.

Print only products whose stock is greater than zero.

Challenge 3: Employee Bonus

Create an `Employee` class with:

```
String name;  
double salary;  
int experience;
```

If experience is 5 years or more, add a 10% bonus.

Print the final salary.

Challenge 4: Objects In An Array

Create a `Student` array that can store three `Student` objects.

Create and place three objects in the array.

Use a loop to print every student's name and marks.

Hint:

```
Student[] students = new Student[3];
```

Part G: Oral Revision

1. When should we use `for` instead of `while` ?
2. Why does `do-while` execute at least once?
3. What causes an infinite loop?
4. What is the difference between `break` and `continue` ?
5. What is the role of the outer and inner loop?
6. What is the difference between a class and an object?
7. Which statement creates an object?
8. How do we access a field from another class?
9. Can one class create multiple objects?
10. Does each object have separate instance-variable values?

Submission Checklist

- Every loop eventually stops.
- Correct update operator is used.
- No unwanted semicolon appears after a loop.
- Class names use PascalCase.
- Objects are created using `new` .
- Fields are accessed using `reference.field` .
- Different objects contain appropriate values.
- Program output is written clearly.

